# PAWAN ALLEM

**Applied AI Engineer • AgenticAI Systems Engineer • LLM Engineer**
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## EXECUTIVE SUMMARY

Applied AI Engineer with 4+ years of experience building production AI systems and evolving into Agentic AI engineering. Experienced in designing local-first AI platforms, intelligent developer tools, and autonomous workflows using LangGraph, LangChain, MCP, Ollama, FastAPI, and modern LLMs. Currently architecting an Internal AI Productivity Platform adopted by a 10-member engineering team that automates documentation, code understanding, engineering knowledge retrieval, project planning, terminal operations, report generation, and reusable workflows. My engineering philosophy is to build reliable AI systems that reason, retrieve, plan, and execute real work—not isolated demonstrations.

## SELECTED IMPACT

• 4+ years delivering production AI solutions for enterprise and government organizations.
• Designed a local-first AI platform used daily by a 10-member engineering team.
• Strong software engineering foundation combined with modern Agentic AI and LLM orchestration.
• Building reusable AI systems that improve developer productivity through autonomous workflows.

## CORE EXPERTISE

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| **Agentic AI** | LangGraph, LangChain, AI Workflow Automation, MCP, Tool Calling, Deep Research Agents, Prompt Engineering |
| **LLM Engineering** | Gemma 4, Qwen Coder, Ollama, Local LLMs, RAG, Embeddings, Structured Outputs |
| **AI Engineering** | Python, FastAPI, REST APIs, SQL, Docker, Git, Modular Architecture |
| **Machine Learning** | PyTorch, TensorFlow, TensorFlow Lite, Computer Vision, OCR, YOLO, Scikit-learn |
| **Focus Areas** | Developer Productivity, Autonomous Research Systems, AI Automation, Intelligent Workflows |

## PROFESSIONAL EXPERIENCE

**Machine Learning Engineer (Applied AI) | Vassar Labs | May 2022 – Present**

- Architected and delivered production AI systems across intelligent automation, computer vision, edge AI, and enterprise software.
- Designed and built an Internal AI Productivity Platform using LangGraph, LangChain, Ollama, FastAPI, MCP, Gemma 4, and Qwen Coder to automate documentation, repository exploration, code explanation, project planning, report generation, terminal operations, and engineering workflows.
- Engineered a secure local-first AI architecture where routine engineering work is executed by local LLMs while Claude Code is leveraged for architecture, complex implementation, and higher-order reasoning.
- Collaborated directly with engineering leadership to identify repetitive knowledge work and transform it into reusable AI capabilities adopted across the team.
- Built modular AI infrastructure that emphasizes extensibility, maintainability, and measurable developer productivity gains.

## SELECTED AI SYSTEMS

- **AI Developer Copilot:** Internal AI assistant that automates documentation, code understanding, file discovery, terminal commands, report generation, project planning, and reusable engineering workflows through a unified dashboard.
- **Autonomous Research Engineer:** CrewAI workflow that searches arXiv, downloads papers, extracts methodology, summarizes implementation details, and generates developer-ready implementation reports using a fully local Gemma 4 model.
- **Deep Research Agent:** LangGraph-based autonomous research system implementing planning, iterative reasoning, retrieval, tool orchestration, and structured report generation inspired by modern Deep Research architectures.
- **Enterprise Knowledge Intelligence Platform:** Production-ready RAG platform using LangChain, embeddings, vector retrieval, and FastAPI for citation-grounded enterprise knowledge discovery.

## EDUCATION, AWARDS & PROFESSIONAL DEVELOPMENT

• B.Tech., Computer Science Engineering — GITAM University
• Oracle AI Foundations Associate
• freeCodeCamp Python Certification
• Best Employee Award — Vassar Labs (2022 & 2023)
• Continuous research in Agentic AI, LangGraph architectures, Deep Research systems, and AI engineering best practices.